

EXP 3 : Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

Code :

```
import pandas as pd

import matplotlib.pyplot as plt

from sklearn.preprocessing import LabelEncoder

from sklearn.tree import DecisionTreeClassifier, plot_tree


dataset = pd.DataFrame({

    "Outlook": ["Sunny","Sunny","Overcast","Rain","Rain","Overcast","Sunny","Rain"],

    "Temperature": ["Hot","Hot","Hot","Mild","Cool","Cool","Mild","Mild"],

    "Humidity": ["High","High","High","High","Normal","Normal","Normal","High"],

    "Wind": ["Weak","Strong","Weak","Weak","Weak","Strong","Strong","Strong"],

    "Play": ["No","No","Yes","Yes","Yes","Yes","Yes","No"]

})


print("Dataset")

print(dataset)


le = LabelEncoder()


for column in dataset.columns:

    dataset[column] = le.fit_transform(dataset[column])


X = dataset.iloc[:, :-1]

y = dataset.iloc[:, -1]
```

```

model = DecisionTreeClassifier(criterion="entropy", random_state=0)
model.fit(X, y)
plt.figure(figsize=(10,6))
plot_tree(
    model,
    feature_names=["Outlook","Temperature","Humidity","Wind"],
    class_names=["No","Yes"],
    filled=True,
    rounded=True
)

plt.title("Decision Tree using ID3 Algorithm")
plt.show()

```

```

sample = pd.DataFrame({
    "Outlook":["Sunny"],
    "Temperature":["Cool"],
    "Humidity":["High"],
    "Wind":["Weak"]
})

```

```

for column in sample.columns:
    encoder = LabelEncoder()
    encoder.fit(["Sunny","Rain","Overcast"] if column=="Outlook"
               else ["Hot","Mild","Cool"] if column=="Temperature"
               else ["High","Normal"] if column=="Humidity"
               else ["Weak","Strong"])
    sample[column] = encoder.transform(sample[column])

```

```
prediction = model.predict(sample)
```

```
if prediction[0] == 1:
```

```
    print("\nPrediction: Play = Yes")
```

```
else:
```

```
    print("\nPrediction: Play = No")
```

Dataset						
	Outlook	Temperature	Humidity	Wind	Play	
0	Sunny	Hot	High	Weak	No	
1	Sunny	Hot	High	Strong	No	
2	Overcast	Hot	High	Weak	Yes	
3	Rain	Mild	High	Weak	Yes	
4	Rain	Cool	Normal	Weak	Yes	
5	Overcast	Cool	Normal	Strong	Yes	
6	Sunny	Mild	Normal	Strong	Yes	
7	Rain	Mild	High	Strong	No	



